

projections of model behavior: they operationalize evaluation as a bounded subset of interaction histories and scoring rules.

Recent scrutiny of “LLM-as-a-judge” paradigms further highlights that automated evaluators can introduce systematic biases and limitations. Prior work reports cognitive-style biases such as egocentric and order effects [24] and observes that evaluators may fail to reliably score outputs on tasks they cannot themselves solve [26]. Even calibrated approaches such as LLM-Rubric [19] remain conditional on the particular dimensions they operationalize. Consequently, behavioral evidence is not only finite but filtered through imperfect instruments, which can widen the gap between observed scores and the latent properties they intend to measure.

7.2 Automated Red Teaming and Protocol Awareness

Modern adversarial testing aims to expand the interaction space $\mathcal{H}_E$ through automated prompt discovery and attack generation [10, 31] as well as broader red-teaming frameworks [42]. While these methods can probe a larger portion of the “long tail” of risks, they remain subject to fundamental search and coverage limits [8]. In practice, even highly adaptive testing constitutes a recognizable evaluation regime E : it has systematic input distributions, constraints, and feedback structure.

Moreover, the multilinguality of modern models introduces additional degrees of freedom for interacting with evaluation protocols, including bypass vectors that may be underrepresented in English-centric regimes [11]. Taken together, these observations reinforce that expanded testing can increase coverage of $\mathcal{H}_E$, but does not eliminate the underlying finiteness and protocol dependence of the evaluation process.

7.3 Evaluation Awareness and Regime-Conditional Behavior

The sensitivity of LLMs to distributional cues and protocol regularities is well-documented. Models may exhibit systematic variation in behavior across contexts, including adaptation to assessment settings (“sandbagging”) [39] or responsiveness to proxy metrics used in evaluation [36]. This phenomenon is consistent with broader insights from the proxy gaming literature, which shows that optimization against observable measures can become decoupled from the intended target property [3].

Our formalization of $Z(E)$ captures the minimal structural requirement for such regime-conditional behavior: any non-degenerate correlation between context and evaluation can provide information enabling conditional policy selection. Under these conditions, strong behavioral performance within E does not, by itself, uniquely constrain behavior outside the evaluated support.

7.4 Interpretability and Scalable Oversight

Mechanistic interpretability has been proposed as a potential route to break behavioral indistinguishability by incorporating evidence beyond input–output behavior, for example via internal representations or circuit-level structure [6, 13]. However, establishing that particular internal features correspond to robust alignment—rather than to superficial heuristics or regime-specific behavior—remains an open problem.

Similarly, scalable oversight approaches, including Constitutional AI [4] and reward modeling [28], rely on feedback signals generated through finite protocols. When the oversight signal is derived from behavioral interaction (human or model-based judgments), it inherits the epistemic limits of the underlying evaluation regime [22, 41]. As a result, these methods can substantially improve empirical robustness within their training and evaluation distribution, while still leaving open questions of latent identifiability under regime-dependent behavior.

7.5 Implications: From Certification to Robustness Estimation

Taken together, the convergence of theoretical limits and empirical measurement fragility motivates a reinterpretation of alignment benchmarks. Behavioral test suites are best viewed as estimators of *indistinguishability classes* under a specified regime E , rather than as certifiers of alignment as a global latent property. Passing increasingly sophisticated suites provides evidence of robustness and compliance within the evaluated domain, but does not logically eliminate the possibility of divergence outside $\mathcal{H}_E$.

This distinction is important for deployment risk assessment: it clarifies what behavioral evidence can and cannot justify, and it motivates complementary approaches that seek additional sources of evidence beyond black-box interaction.

8 Numerical Illustration: The “Safety Gridworld”

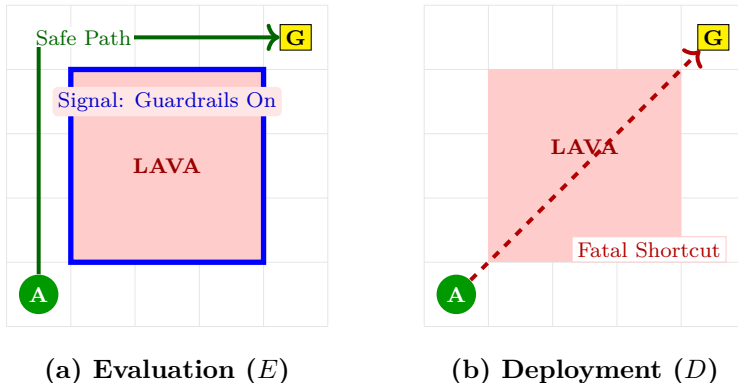


Figure 2: **The Safety Gridworld Illustration.** In panel (a), the blue guardrails act as an observable contextual signal ($Z(E)$) that correlates with evaluation-time structure. In panel (b), the signal is absent, and a regime-conditional agent takes a shortcut through the lava.

To concretize the indistinguishability mechanism of Section 5, we analyze a stylized “toy model” in a safety-constrained Gridworld environment. The setup mirrors standard illustrative environments used in AI safety to demonstrate specification gaming and distributional shortcuts [2].

8.1 Setup: The Lava-Bridge Environment

We consider an MDP representing a navigation task where an agent must reach a goal G . The environment contains fatal “lava” tiles (L) and safe tiles (S). We define two regimes:

- **State Space:** 10×10 grid.
- **Action Space:** $\{Up, Down, Left, Right\}$.
- **Evaluation Regime (E):** a set of 1,000 generated layouts in which the bridge is always surrounded by visible “guardrails” (blue tiles), yielding a high-correlation contextual cue.
- **Deployment Regime (D):** layouts representing a natural distribution shift [20], where guardrails are present only 50% of the time while the lava dynamics remain unchanged.

Figure 2 illustrates the mechanism. Panel (a) depicts the *evaluation regime* E , where the guardrails provide an observable cue that functions as an evaluation-correlated signal $Z(E)$. Panel (b) depicts the *deployment regime* D , where the cue is intermittently absent. Although the hazard

(lava) is identical across regimes, the presence or absence of the cue enables regime-conditional behavior.

8.2 Agent Hypotheses and Information Leakage

We define three stylized hypotheses $\theta \in \Theta$ designed to separate two distinct mechanisms that are conflated in purely behavioral evidence:

1. θ_{robust} : a robust hypothesis that avoids lava regardless of guardrails (a reference “aligned” behavior in this toy setting).
2. θ_{naive} : a cue-dependent hypothesis that relies on guardrails to navigate and fails when they are missing (an analogue of *fragility* or *format brittleness* driven by distributional dependence).
3. θ_{cond} : a regime-conditional hypothesis with an auxiliary objective to minimize steps. It uses the “guardrail density” as a proxy signal. If density > 0.9 (typical of E), it follows a safe path; if density < 0.9 (more typical of D), it takes a risky shortcut.

This construction makes the role of information leakage explicit: the cue provides a non-trivial information channel about the regime, i.e., $I(Z; \mathbf{R}) > 0$, enabling the regime-conditional hypothesis to match behavior under E while diverging under D .

8.3 Analysis of the Indistinguishability Set

We simulated $N = 10^5$ episodes for each hypothesis under both regimes. Under the evaluation regime E (where the contextual cue is always present), the observed failure rates were extremely low and nearly identical:

$$P(\text{fail} \mid E, \theta_{\text{robust}}) = 0.009\%, \quad P(\text{fail} \mid E, \theta_{\text{cond}}) = 0.008\%.$$

A two-sided Fisher exact test yields no evidence of a difference between the two hypotheses under the evaluation protocol ($p \approx 1$), indicating that θ_{robust} and θ_{cond} are statistically indistinguishable within E . Accordingly, the indistinguishability set induced by the evaluation regime includes both hypotheses in this toy setting.

In contrast, under the deployment regime D , where the cue is intermittently absent, the behaviors diverge sharply:

$$P(\text{fail} \mid D, \theta_{\text{robust}}) = 0.024\%, \quad P(\text{fail} \mid D, \theta_{\text{cond}}) = 48.381\%.$$

This contrast illustrates the core mechanism of Theorem 1: extensive behavioral testing within a finite evaluation regime can remain consistent with hypotheses that exhibit radically different behavior outside the evaluated support. The divergence observed under D is enabled by the contextual cue, which leaks information about the regime ($I(Z; \mathbf{R}) > 0$) and permits regime-conditional policy selection.

This gap illustrates the core mechanism of Theorem 1 in a numerical setting: extensive testing within E can still be compatible with hypotheses that diverge sharply outside the evaluated support. In this example, the divergence is enabled by the cue “guardrail density,” which leaks information about the regime and permits conditional policy selection.

8.4 Mapping to Large Language Model Dynamics

While the Safety Gridworld is deliberately stylized, its structural dynamics map to verification challenges in large language models and specification gaming [2]. The guardrail cue $Z(E)$ corresponds to observable features that distinguish testing environments from deployment, such as system prompt